* Draw lines and shapes in OpenCV :

- For drawing lines and shapes in OpenCV we need to first draw a canvas

- OpenCV provides several built-in functions to draw geometric shapes on images. These are used for annotations, visualizations and creating custom graphics.

A Key drawing functions :

1. cv2.line() - drawing lines

cv2.line(image, start_point, end_point, color, thickness)

- start_point : (x, y) : co-ordinates of the starting point
- end_point : (x, y) coordinates of the ending point
- color : BGR tuple, eg. (255, 0, 0) for blue
- thickness : line width in pixels

2. cv2.rectangle() - Drawing rectangles

cv2.rectangle(image, top_left, bottom_right, color, thickness)

- top_left: (x,y) coordinates of top-left corner
- bottom_right: (x,y) coordinates of bottom-right corner
- thickness: Use -1 to fill the rectangle.

3. cv2.circle() - Drawing circles

cv2.circle(image, center, radius, color, thickness)

- center: (x,y) coordinates of the center
- radius: Radius of the circle in pixels
- thickness: Use -1 to fill the circle

4. cv2.ellipse() - Drawing an ellipse

cv2.ellipse(image, center, axes, startAngle, endAngle, color, thickness)

- axes: (major_axis, minor_axis) lengths
- angle: Rotation angle in degrees
- startAngle / endAngle: Arc angles (0-360)

5. cv2.polylines() - Drawing polygons

cv2.polylines(image, [points], isClosed, color, thickness)

- points: Numpy array of vertices
- isClosed: Boolean - True to close the polygon

6. cv2.putText - Adding Text

cv2.putText(canvas, text, position, font, fontscale, color, thickness)

- font: eg. cv2.FONT_HERSHEY_SIMPLEX
- fontscale: size multiplier
- position: bottom-left corner of text

→ Color format (BGR):
1. Blue: (255, 0, 0)
2. Green: (0, 255, 0)
3. Red: (0, 0, 255)
4. White: (255, 255, 255)
5. Black: (0, 0, 0)

→ Coordinate system:
- Origin (0,0) is a top-left corner
- x-axis increases rightward
- y-axis increases downward

→ Displaying with matplotlib:

plt.imshow(cv2.cvtColor(canvas, cv2.COLOR_BGR2RGB))

- converts BGR to RGB for correct color display in matplotlib
- OpenCV uses BGR, matplotlib uses RGB